

ELP-101 Project Report

Battery Level Indicator

Measuring the signal level of an input making use of LM3914 IC

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TABLE OF CONTENTS

S. No.	Content	Page No.
1.	Aim	3
2.	Apparatus Required	3
3.	Circuit Diagram	3
4.	Theory	4
5.	Working Principle	6
6.	Observations and Verification	7
7.	Real Life Applications	8
8.	Precautions	8
9.	Final Circuit	9

AIM:

To design a LED based battery level indicator that makes use of a LM3914 IC.

APPARATUS REQUIRED:

- Integrated Circuit: LM3914 IC
- Battery
- 10 LEDs
- Resistors: $4.7\text{ k}\Omega$, $18\text{ k}\Omega$, $56\text{ k}\Omega$
- Potentiometer (Analog): $10\text{ k}\Omega$
- Breadboard
- Connecting Wires

CIRCUIT DIAGRAM:

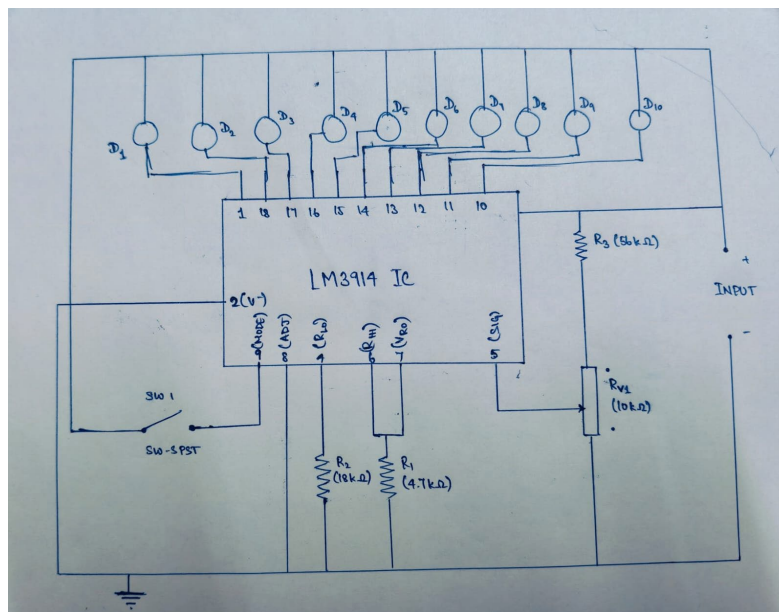


Fig. 1(Circuit Diagram of Battery Level Indicator)

THEORY:

- Working of LM3914 IC:

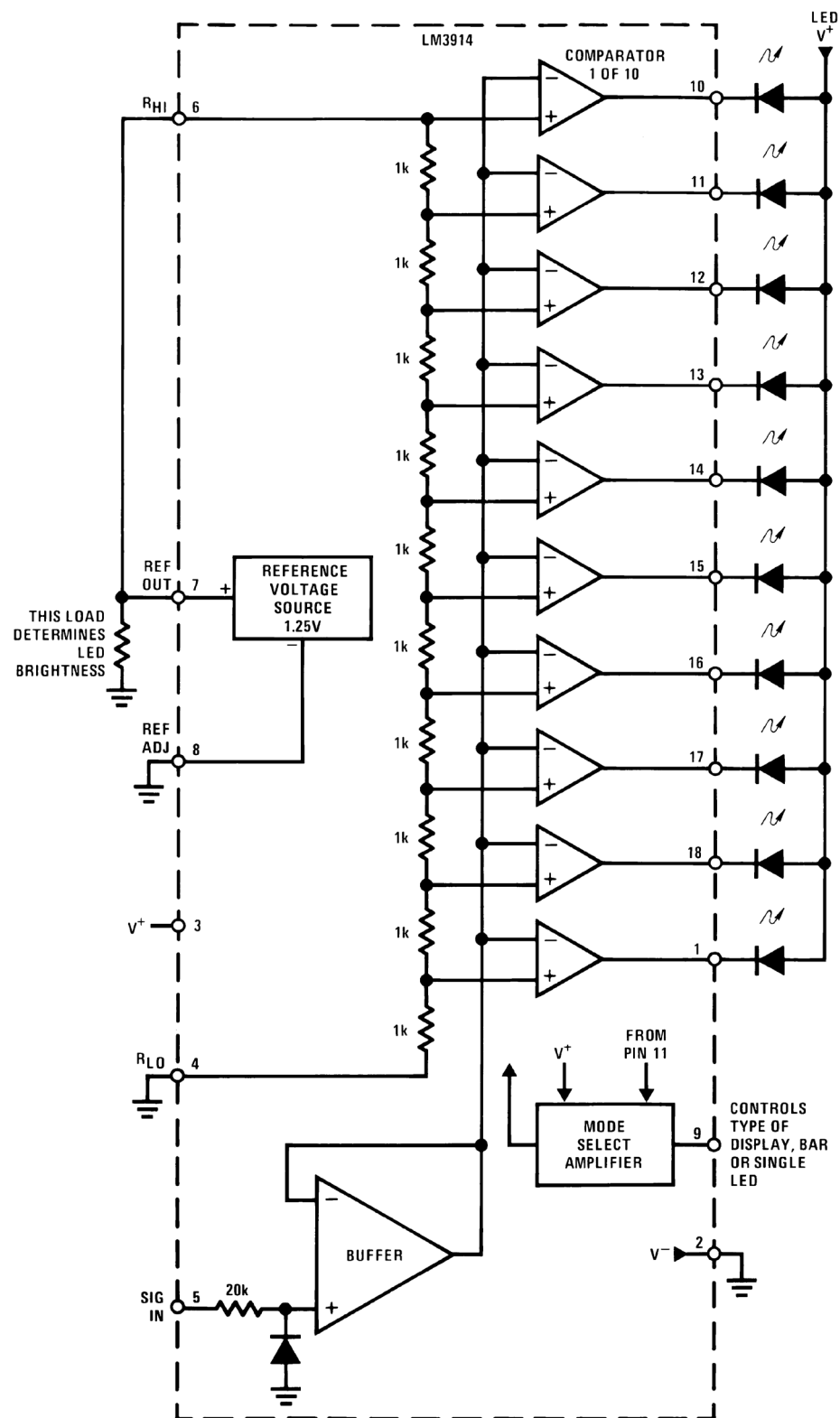


Fig. 2. (Inner Schematic of the LM3914 IC)

The following is the description of the 18 pins present in the LM3914 IC:

- PIN 1: LED No.1
- PIN 2: V_+ (Positive Potential)
- PIN 3: V_- (GND Potential)
- PIN 4: R_{LO} (Lower Reference Potential)
- PIN 5: Signal In
- PIN 6: R_{HI} (Higher Reference Potential)
- PIN 7: R_{OUT} (Output Reference Potential (After changing R_{HI} to different value))
- PIN 8: REF_{ADJ} (Pin used to adjust the R_{OUT} making use of external resistors)
- PIN 9: MODE (To change between Bar and Dot Mode)
- PIN 10: LED No.10
- PIN 11: LED No.9
- PIN 12: LED No.8
- PIN 13: LED No.7
- PIN 14: LED No.6
- PIN 15: LED No.5
- PIN 16: LED No.4
- PIN 17: LED No.3
- PIN 18: LED No.2

Basic explanation on how the LM3914 IC works:

The LM 3914 IC consists of 10 comparators all receiving signals in its negative terminal from PIN No. 5(Signal In) and to its positive terminal from a series of equally spaced potential values which are obtained making use of a series of $1k\Omega$ resistors. (Refer Fig.2)

Each LED lights up when the Signal Input is greater than its Reference voltage in the positive terminal since the Output given in such a case becomes zero and thus causes the LED to be forward biased.

The signal is run through a buffer as shown in Fig. 2 to protect the IC and the diodes from surge protection and also reverse potentials.

The Mode select amplifier connected with Pin No.9 makes use a Logic system and to transistors to give the corresponding out in the dot form where the output is given in such a way that only one LED is lit up at a time with a small buffer voltage where the preceding LED is also lit up.

The Reference voltage is by default 1.25V as setup in the IC but this can be altered making use of the following circuit.

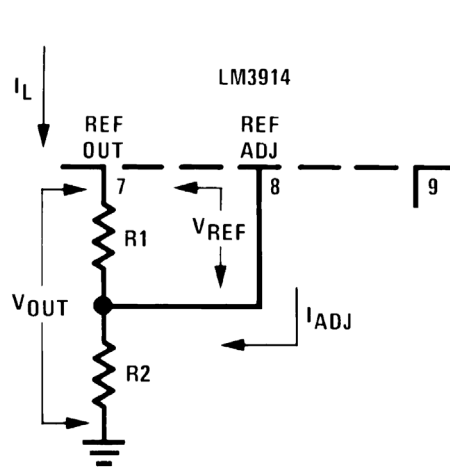


Fig. 3. (Changing V_{REF} using external resistors)

The following Equation is obeyed:

$$V_{OUT} = V_{REF} \left(\frac{R_2}{R_1} + 1 \right) + I_{ADJ} R_2$$

WORKING PRINCIPLE:

From Fig. 2 we can find the V_+ values at which each of the LED turns on. This is because the values of V_+ acts as our battery level which we are meant to indicate.

$$V_{SIG} = V_{IN} \frac{R_V}{R_V + R_3} \left(\frac{\theta_{calibration}}{\theta_{max}} \right)$$

$\theta_{calibration} \rightarrow$ Set by user by assigning maximum value of the voltage to be read

$$V_{Ref,LED_n} = 1.25 \frac{(R_2(in\ k\Omega) + n)}{10 + R_2(in\ k\Omega)}$$

$n \rightarrow$ Serial No. of the LED

$\theta_{calibration}$ is identified by using the case where the signal given through Pin No.5 is just equal to the Reference voltage of the 10th LED.

This gives:

$$\theta_{calibration} = \left(\frac{1.25}{V_{max}} \right) \left(\frac{R_V + R_3}{R_V} \right) \theta_{max}$$

Now we make can make use of this value to find the range of the battery indicator that we have made and the Stepping value for each of the LEDs turning on.

$$V_{min} = V_{max} \frac{(R_2(in\ k\Omega) + 1)}{10 + R_2(in\ k\Omega)}$$

$$V_{step} = \frac{V_{max}}{10 + R_2(in\ k\Omega)}$$

Thus the LEDs light up one by one increments of $\frac{V_{max}}{10 + R_2(in\ k\Omega)}$, starting from a value of $V_{max} \frac{(R_2(in\ k\Omega) + 1)}{10 + R_2(in\ k\Omega)}$ up until V_{max} .

Substituting our resistance values from the circuit constructed we get that

$$V_{min} \cong 0.68 V_{max}$$

$$V_{step} \cong 0.035 V_{max}$$

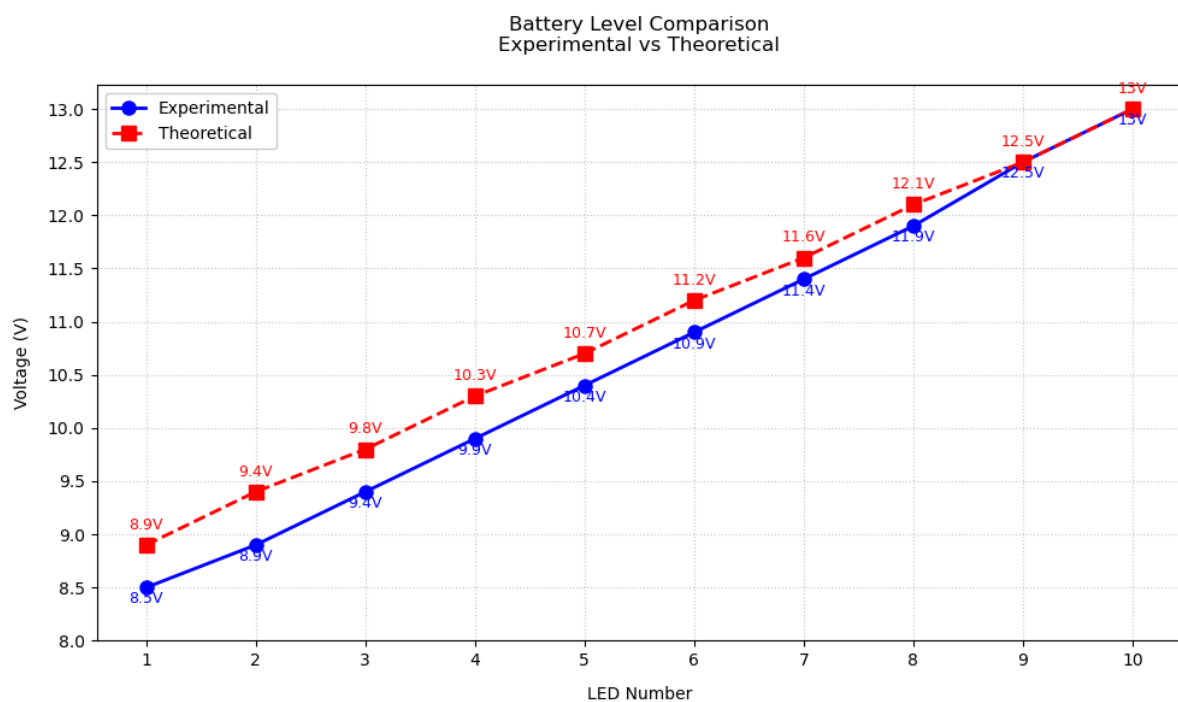
Significance of the resistors used:

- R_1 regulates the current flowing through each of the LEDs.
- R_2 shifts the range of the battery towards the more desirable upper half of the battery's level.
- R_3 is used as a shunt to ensure that only a small part of the voltage that is given by the Battery in test is being passed into the IC. This ensures that the IC doesn't get fried upon usage.

OBSERVATIONS AND VERIFICATION:

Table. No. 1(Experimental vs Theoretical Values)

LED No.	Experimentally obtained Battery Level (in Volts)	Theoretically Expected Battery Level (in Volts)
1	8.5	8.9
2	8.9	9.4
3	9.4	9.8
4	9.9	10.3
5	10.4	10.7
6	10.9	11.2
7	11.4	11.6
8	11.9	12.1
9	12.5	12.5
10	13	13



Graph No. 1(Experimental vs Theoretical Values)

REAL LIFE APPLICATIONS:

- As indicators in electronical appliances such and power banks, torches and electrical vehicles etc.
- As indicators of level of various other physical factors such as humidity, temperature and sound which can make use of sensors to give a particular voltage output according to its intensity which can then be used in the above mechanism with proper calibration of the resistor values to our desire.
- In uninterruptible power supplies (UPS systems), LED bar graph indicators are can be used to show both the real-time battery charge level and the power load connected to the system.

PRECAUTIONS:

- Ensure that the values of voltage used across any of the components in the circuit doesn't exceed a certain value.(The LM3914 IC used in this circuit can withhold upto 35V)
- Do not keep the circuit operational for large amounts of time to ensure that all the components and circuit components behave ideally.
- Avoid short circuits when wiring on the breadboard since this can overheat components or damage your power supply or battery.
- Power off the circuit when modifying connections.

FINAL CIRCUIT:

